Înmulțirea 1011 * 1101

1011 * 1101

M	C	A	Q	
1011	0	0000	1101	$Q_0 = 1$
	0	1011	→	
	0	0101	1110	$Q_0 \neq 1$
	0	0010	1111 →	$Q_0 = 1$
	0	1101		
	0	0110	1111 →	
	1	0001		
		1000	1111	$Q_0 = 1$ →

Împărțirea 10010011 / 1011

M	C	A	Q	
1011		1001	0011	
	1	0010	0110	$\leftarrow$ $CA \geq M$, $Q_0 = 1$
	0	0111	0111	
	0	1110	1110	$\leftarrow$ $CA \geq M$, $Q_0 = 1$
	0	0011	1111	
	0	0111	1110	$\leftarrow$ $CA < M$, $Q_0 = 0$
	0	1111	1100	$\leftarrow$ $CA \geq M$, $Q_0 = 1$
	0	0100	1101	

$$\begin{array}{r} 10010 - \\ 1011 \\ \hline 111 \end{array}$$

$$\begin{array}{r} 01110 - \\ 1011 \\ \hline 00011 \end{array}$$

$$\begin{array}{r} 1111 - \\ 1011 \\ \hline 0100 \end{array}$$

$x = 36_{(10)} = 100100_{(2)}$

$[x]_D = \underbrace{0010\ 0100}$

$[x]_J = 0010\ 0100$

$[x]_C = 0010\ 0100$

$y = -72_{(10)} = 1001000_{(2)}$

$72 = 36 = 18 = 9 = 4 = 2 = 1$

0001001

$[y]_D = 11001000$

$[y]_I = 10110111$

$[y]_C = 10111000$

$[x+y]_C = [x]_C \oplus [y]_C$

$[x]_C \ominus [y]_C = [x]_C \oplus [-y]_C$

$[x]_C + [y]_C = 11011100$

$$\begin{array}{r} 00100100 + \\ 10111000 \\ \hline 11011100 \end{array}$$

$[y]_C + [y]_C =$

$$\begin{array}{r} 10111000 + \\ 10111000 \\ \hline 101110000 \\ 100000000 \\ \hline 01110000 \end{array}$$

$101110000 - \boxed{> 2^n}$

01110000 — rezultat

avem depășire (s-au adunat 2 nr. negative și s-a obținut unul pozitiv)

$0,127_{(10)} = 0,0010000_{(2)}$

$$\begin{array}{r} 127 \cdot \\ 2 \\ \hline 254 \cdot \\ 2 \\ \hline 508 \cdot \\ 2 \\ \hline 1016 \cdot \\ 2 \\ \hline 0,032 \cdot \\ 2 \\ \hline 0,064 \cdot \\ 2 \\ \hline 0,128 \end{array}$$

$[x]_D = 00010000$

$[x]_I = 00010000$

$[x]_C = 00010000$

$237,5_{(10)} =$

$237 : 8 = 29 = 3 = 0$

$5 \quad 5 \quad 3$

$237_{(10)} = 355_{(8)} = 011 \ 101 \ 101,1$

$\underset{S}{0} \, \underbrace{110\,1101}_{I} \underset{;}{1} \underbrace{1000\,00001}_{F}$

p. întreagă p. fracționară

$x = 237,5_{(10)} = 11 \ 101 \ 101,1_{(2)} =$

$= 0,111011011_{(2)} * 2^8$

$C = 8 + 127$

$$= \begin{array}{r} 1000 + \\ 1111111 \\ \hline 10000111 \end{array}$$

$\underbrace{0}\ \underbrace{10000111}\ \underbrace{1111011011100\ 0000\ 0000\ 0\ldots}$